Accuracy, Support, and Misspecification in Regression GATE Estimation

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Abstract

Regression GATE estimators predict outcomes under all-treated and all-control assignments from a mixed network experiment. We study their accuracy through conditional bias–variance and estimability diagnostics, using 102 simulation cells with 36 graphs, eight response mechanisms, two assignment designs and 400 repetitions per cell. Correctly matched linear regression tracks endpoint-contrast variance: at 2000 nodes, mean graph RMSE rises from 0.101 to 0.412 as regular degree increases from 4 to 80. Better experimental-distribution prediction, correct exposure features and increased support each have measured counterexamples to guaranteed risk improvement. A coupled hidden-tail response demonstrates a support obstruction, while correlated noise reduces working-independence coverage to about 0.59 on the fixed public topology. Exact full-exposure HT risk also exposes extreme tails missed by finite simulation. These are conditional findings for specified synthetic outcomes and tested topology/design regimes, with graph and Monte Carlo uncertainty distinguished; they introduce no new estimator or feasible oracle inference procedure.

1 Introduction

An intervention on a network can change the outcomes of both treated units and their untreated neighbors. The effect relevant to a platform-wide rollout is therefore different from an own-treatment comparison within a mixed experiment. We study the global average treatment effect (GATE): the average outcome under treating every unit minus the average outcome under treating none. Random assignment identifies the experimental assignment distribution, but estimating this global contrast also requires handling interference between units.

Regression adjustment offers a direct approach: construct features from the assignment and graph, fit an outcome model, and compare its predictions at the two global assignments [Chin, 2019, Han and Ugander, 2023]. The challenge is that fitting outcomes in a mixed experiment and predicting global endpoints are different tasks. Neighbor-treatment fractions can be concentrated away from both endpoints, so a fitted response is used outside the densely observed region. Such a distribution shift and its implications for extrapolation are already explicit in Chin [2019]; regression adjustment and endpoint prediction are not contributions introduced here.

Our organizing question is how to separate uncertainty due to the experimental design from error due to a response or exposure model that fails at the target intervention. For a fixed linear smoother, conditional mean-squared error separates exactly into squared bias and noise variance. This elementary decomposition motivates evaluating endpoint contrast weights together with response misspecification, rather than treating predictive fit, network size, or the number of observed units as sufficient evidence of accurate GATE estimation. Graph-based assignment can change both exposure variation and the number of independently assigned groups, so its net effect must be assessed rather than assumed.

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We distinguish three levels of evidence throughout: algebraic identities conditional on a graph and assignment; repeated-assignment performance on a fixed graph with synthetic outcomes; and variation across generated graph realizations. A public social-network topology supplies a realistic connectivity pattern, but synthetic outcomes on that topology do not establish accuracy for actual platform responses. The intended scope is a controlled assessment of specified regression families and interference mechanisms, with failures and undefined estimates retained in the evaluation.

We test four prespecified questions about matched-model precision, prediction versus endpoint accuracy, allocation/support tradeoffs and mapping/covariance failures (Section 5). The completed study supports conditional accuracy regimes and counterexamples rather than a universal ranking: variance explains matched-regression precision in controlled degree/size slices, but response/mapping bias, unsupported contrasts and covariance uncertainty create distinct failures. Section 6 reports favorable and adverse evidence, including finite-run underrepresentation of rare HT risk. These empirical diagnostics complement the existing regression framework; regression endpoint prediction itself is not a new method.

2 Related work

Exposure mappings and randomization. Aronow and Samii [2017] organize causal inference under interference around an assignment design, a mapping from assignments to unit exposures, and a target estimand. Their inverse-probability estimators and randomization-based variance calculations account for dependence induced by the exposure structure. This framework makes clear why a weighting comparator requires the exposure probabilities of its actual design and an exposure mapping sufficient for the target outcomes. A one-hop exposure condition cannot be assumed valid for arbitrary longer-range interference.

Design and analysis under network interference. Eckles et al. [2017] study both correlated assignment through graph clusters and analyses incorporating neighbors' treatment assignments. Their theoretical conditions and simulations establish that bias reduction depends on network structure and the interaction mechanism. Thus neither the use of graph structure nor a conditional benefit from clustering is new to the present study. The relevant comparison is the resulting balance of bias, variability, and exposure support under a specified design and graph.

Regression prediction of global interventions. Chin [2019] introduce regression adjustment for global treatment effects using assignment-derived features, discuss differences between observed and global feature distributions, and derive linear-model unbiasedness and variance results under explicit assumptions. Their finite-sample formulas already involve inverse feature Gram matrices, and their rank caveat identifies singular feature designs as a practical concern. Han and Ugander [2023] extend the regression approach with sequentially generated and selected graph- and treatment-based covariates and block-bootstrap inference. Fixed low-dimensional regressions therefore represent a restricted comparison class; their failure does not establish failure of all regression adjustment or adaptive feature construction. Our conditional error identities are diagnostic tools within this existing framework, rather than claims to a new regression estimator or a general inference procedure.

3 Target and error diagnostics

For a fixed graph G on n units and assignment $z \in \{0,1\}^n$, let $m_i(z)$ be the conditional outcome mean. We target

$$\tau_G = n^{-1} \sum_i \{m_i(\mathbf{1}) - m_i(\mathbf{0})\}. \tag{1}$$

Observed outcomes are $Y = m(z) + \varepsilon$ with conditional mean-zero noise and covariance Σ. Across repetitions we redraw assignment and noise, holding graph, covariates and partition fixed. Equivalently, attaching the same additive disturbance to each unit under all potential assignments makes it cancel from the global contrast; independently redrawing endpoint noise would define a different target.

For features $X(z)$ fixed conditional on assignment, including basis/tuning choices, minimum-norm OLS gives

$$c_G = n^{-1} \sum_i \{X_i(\mathbf{1}) - X_i(\mathbf{0})\}, \qquad \widehat{\tau} = c_G^\top X(z)^+ Y = a(z)^\top Y. \tag{2}$$

Here X^+ is the Moore–Penrose pseudoinverse and $a = (X^+)^\top c_G$. Define $b(z) = a^\top m(z) - \tau_G$ and $v(z) = a^\top \Sigma a$. Then

$$\mathbb{E}[(\widehat{\tau} - \tau_G)^2 \mid z, G] = b(z)^2 + v(z), \qquad \text{MSE} = \mathbb{E}_z[b(z)^2] + \mathbb{E}_z[v(z)]. \tag{3}$$

Bias is $\mathbb{E}_z[b(z)]$ and variance is $\text{Var}_z\{b(z)\} + \mathbb{E}_z[v(z)]$. Independent homoscedastic noise gives $v = \sigma^2 \|a\|^2$; a known construction $\varepsilon = Bu$, $\text{Cov}(u) = I$, gives $v = \|B^\top a\|^2$. These are explanatory identities, not new inference theorems. Their proof is immediate from $\widehat{\tau} - \tau_G = b + a^\top \varepsilon$: mean-zero noise removes the squared-error cross term, and total variance gives the marginal decomposition. No independence of node exposures is used. Evaluating b, v requires the mean/covariance known in simulation; a small weight norm cannot certify small unknown misspecification bias. Outcome-adaptive feature selection is outside this calculation.

If common coefficients describe both experimental and endpoint means, the contrast is reproduced when $c_G \in \text{row}(X)$. Indeed X^+X projects onto that row space, so $a^\top X = c_G^\top$. We flag estimability by $\|c_G - \Pi_{\text{row}(X)} c_G\| < 10^{-8}$. Rank deficiency need not prevent an estimable contrast, but a finite pseudoinverse return for a nonestimable contrast is an extrapolation convention, not identified recovery.

4 Simulation framework

Outcomes, exposure and noise. Let A be simple undirected adjacency, with degrees d_i, and P its row-normalization, replacing isolate rows by identity rows. One-hop exposure is the treated-neighbor fraction $s = Pz$, computed by counts followed by division; isolates use own treatment. Two-hop exposure is $t = Ps$, including random-walk backtracking. Both global assignments therefore have exposure endpoints zero/one. For fixed $x_i \sim N(0, 1)$ independent of graph/assignment,

$$m_i(z) = 0.5x_i + z_i + \gamma h_i(z), \qquad \gamma = 1. \tag{4}$$

Eight responses test matched adjustment, smooth/asymmetric misspecification, abrupt support failure, omitted range and hidden endpoint curvature: no interference $h_i = 0$; s_i, s_i^2, s_i^3; an endpoint-normalized sigmoid; $\mathbf{1}\{s_i \geq 0.75\}$; $(s_i + t_i)/2$; and

$$h_i = s_i + \{\max((s_i - 0.85)/0.15, 0)\}^2. \tag{5}$$

The last mechanism matches the linear response on $[0, 0.85]$ but differs at rollout. Truth is computed by the generator at both global assignments: analytically 1 without interference, $1 + 2\gamma$ for the hidden tail, and $1 + \gamma$ otherwise. Noise is $\varepsilon = B_\rho u$, $u \sim N(0, I)$, where each row of $I + \rho P$ is normalized to unit Euclidean norm. Every unit has marginal variance one; $\rho = 0$ gives independent errors and $\rho = 1$ a specified local covariance. Exact response/basis/numerical details are in Appendix B.

Assignment and comparisons. We use synthetic regular, Erdős–Rényi, block-model and preferential-attachment graphs plus the full SNAP combined ego-Facebook topology [Stanford Network Analysis Project, McAuley and Leskovec, 2012]. Public topology supplies no real experimental outcomes or population sample. Preprocessing removes loops/duplicates to obtain simple undirected adjacency; acquisition is hash-verified. A seeded random node order and breadth-first search form connected partition pieces of at most 32 nodes, retaining smaller remainders without using outcomes or planted labels. Individual assignment treats nodes independently with probability p; cluster assignment treats these fixed pieces independently with probability p. Covariates/partitions remain fixed across repetitions. Table 1 records measured topology/partition properties.

DIM fits intercept and own treatment, equaling the ordinary treated-minus-control mean when both arms exist. Other regressions also include x: linear/quadratic/cubic powers of s, a fixed cubic spline, a two-hop model with s, t, and a true-cutoff threshold-feature oracle. No basis is chosen using outcomes or endpoint truth. The threshold oracle diagnoses known mechanism specification, not cutoff discovery. All fits use SVD minimum-norm OLS and retain the estimability flag.

For full-neighborhood HT/Hájek baselines, E_i^a requires own and every immediate neighbor treatment equal a. If k_i counts independent assignment units in this closed neighborhood, $\pi_i^1 = p^{k_i}$ and $\pi_i^0 = (1-p)^{k_i}$; $k_i = d_i + 1$ under node assignment and counts distinct partition pieces under clusters. HT averages $\mathbf{1}\{E_i^1\}Y_i/\pi_i^1 - \mathbf{1}\{E_i^0\}Y_i/\pi_i^0$ over nodes; Hájek normalizes each exposure arm by its realized weight total. This mapping reveals the one-hop endpoints but can be insufficient for two-hop outcomes. Exact joint-exposure risk (Appendix A.3) checks rare-event contributions missed by finite simulation.

Each training assignment has an independent fresh assignment/noise draw for prediction evaluation, unavailable to the estimator. Saved prediction errors distinguish fresh noisy outcomes, conditional means and training residuals. This evaluates the experimental allocation distribution, not global rollout. Per-fit diagnostics retain conditional bias, exact noise variance, support, estimability, and Gaussian known-covariance oracle and working-independence HC3 intervals. Neither interval removes misspecification bias; oracle covariance is unavailable in practice. DIM has intervals only when both arms exist; HT/Hájek have no implemented interval claim.

5 Evaluation protocol

Prespecified questions and grid. H1 tests degree/size versus endpoint-contrast variance in matched models. H2 compares experimental-distribution prediction with GATE error, using quadratic complement cancellation and a hidden-tail coupling control. H3 asks when cluster exposure support compensates for changed assignment units. H4 tests omitted two-hop exposure, correlated noise, unsupported contrasts and rare full exposure.

Each of 102 cells uses 400 assignment/noise repetitions, eight mechanisms and nine estimators. The 72-cell core crosses both designs at $p = 0.5, \rho = 0$ with 35 synthetic graphs and one public topology. Seven generated regimes, each with seeds 201–205, are regular $n = 2000$, degrees 4/20/80; regular $n = 8000$, degree 20; and Erdős–Rényi, eight-block and preferential-attachment $n = 2000$, target degree 20. Regular slices isolate degree or size interventions; cross-family contrasts change multiple properties. The 22-cell $\rho = 1$ panel repeats degree-20 regular, block-model and SNAP cases. Eight imbalance cells use only the seed-201 degree-20 regular graph and SNAP, both designs, $p = 0.2/0.8, \rho = 0$. This is a focused sensitivity check. All use $\gamma = 1$. Masters 2202601/2202602/2202603 separate core/noise/imbalance streams from pilot 89103. All 102 corrected cells completed; Appendix C documents source/graph/configuration checks, the rational-boundary repair and clean selected replay.

Error, failures and uncertainty. For $e_r = \widehat{\tau}_r - \tau_G$, signed bias measures systematic displacement, sample SD spread, and $\mathrm{RMSE} = \sqrt{R^{-1}\sum_r e_r^2}$ their combined outcome-scale error. We compare MSE with averages of $b(z)^2$ and $v(z)$, not only squared average bias. Operational regression risk retains every finite minimum-norm return, with nonestimable fractions and separate estimable-only denominators. DIM is undefined for an empty arm; Hájek is undefined if either exposure-arm weight total is zero. Their finite-output risk is conditional on a defined return, accompanied by failure fractions; no artificial error is assigned to missing output. These conditional and unconditional risks cannot be ranked indiscriminately.

Fixed-graph bias MCSE is $s(e)/\sqrt{R}$, MSE MCSE $s(e^2)/\sqrt{R}$, and RMSE MCSE the latter divided by twice RMSE. Normal/delta intervals are approximate and unreliable for unobserved extreme tails. Coverage/failure proportions use Wilson intervals. Models/mechanisms within a cell share assignment/noise, supporting paired contrasts. Different designs and production panels have independent replicate streams; matching graph labels does not create replicate pairing.

Synthetic summaries weight graphs equally and use t_4 intervals across five graph summaries, including their finite MC variation. Pooled RMSE (root mean graph MSE) differs from mean graph RMSE. SNAP and one-seed imbalance yield conditional MC uncertainty, not graph-population intervals. Separately, empirical coverage assesses nominal 95% single-experiment intervals, retaining finite-interval and estimable-only denominators alongside failures/nonestimability. HC3 assumes working independence; cluster assignment alone does not imply correlated noise, and correlated-noise oracle intervals are simulation diagnostics. All-success/failure empirical fractions and degenerate graph intervals do not establish probabilities zero/one.

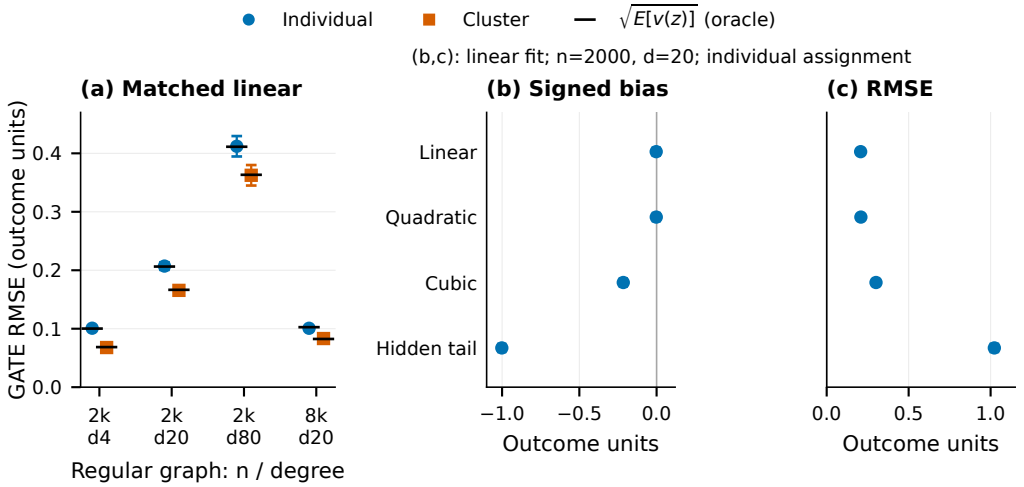


Figure 1: Accuracy depends on degree, size and response shape. Regular graphs; $p = 0.5$, $\rho = 0$, $\gamma = 1$; five independent graph seeds, 400 assignment/noise attempts per graph. Points are equal-graph means and bars are supplied approximate 95% t_4 intervals across graphs, including finite Monte Carlo variation; no multiplicity adjustment. (a) Correctly specified linear regression on the linear response under individual/cluster assignment; labels give n (2k/8k) and regular degree d. Black ticks are $\sqrt{E[v(z)]}$, the simulation-known noise contribution after averaging conditional variance, with no interval; this differs from mean graph RMSE. (b,c) Linear regression under individual assignment at $n = 2000, d = 20$ on four responses. Bias and RMSE use outcome units; all 400 outputs per graph are defined and endpoint-estimable. Near-zero quadratic signed bias at half allocation does not imply low RMSE or absent conditional bias. Table 2 retains the numerical values and squared-bias contribution. Noise/imbalance panels are outside this core display.

6 Results

All 102 cells completed: 40,800 assignment/noise repetitions, 2,937,600 estimator outputs and 306 exact one-hop HT risks. Below, intervals are approximate 95% t_4 intervals over five equally weighted graph summaries unless identified as fixed-graph Monte Carlo (MC) uncertainty. Graph intervals include finite simulation variation. These explanatory comparisons answer H1–H4; they are not multiplicity-adjusted discovery tests or a universal ranking. Figures 1–3 show precision, prediction and stress comparisons; Appendix D retains complete numerical companions, including all topology, failure and exact-risk rows. Complete source rows, denominators and uncertainty routes are registered in `results/diagnostics/claim_evidence.csv`.

Matched precision depends on degree as well as size (H1). Under individual assignment and the linear response, all matched linear contrasts are estimable. At $n = 2000$, mean graph RMSE increases from 0.1009 $[0.09888, 0.1029]$ at regular degree 4, to 0.2072 $[0.2007, 0.2136]$ at degree 20, and 0.4121 $[0.3946, 0.4295]$ at degree 80. At degree 20, increasing size to $n = 8000$ reduces RMSE to 0.1008 $[0.09779, 0.1037]$. Mean oracle variances are respectively 0.01005, 0.04257, 0.1691 and 0.01053, consistent with noise amplification through the endpoint contrast. Mean graph RMSE and the square root of mean variance have different averaging orders. Their agreement is a diagnostic within a matched regime, not a feasible check for unknown response bias or a general graph-asymptotic rate.

Better prediction can worsen GATE; cancellation is allocation-specific (H2). On regular $n = 2000$, degree-20 graphs with cluster assignment and cubic outcomes, cubic-minus-linear fresh-outcome prediction MSE is -0.001980 $[-0.002273, -0.001686]$, whereas its GATE-MSE difference is $+0.1324$ $[0.1054, 0.1594]$. Both fits are fully defined and estimable; shared assignments/noise are retained in these paired comparisons. Better prediction under the experimental allocation distribution does not certify lower endpoint risk.

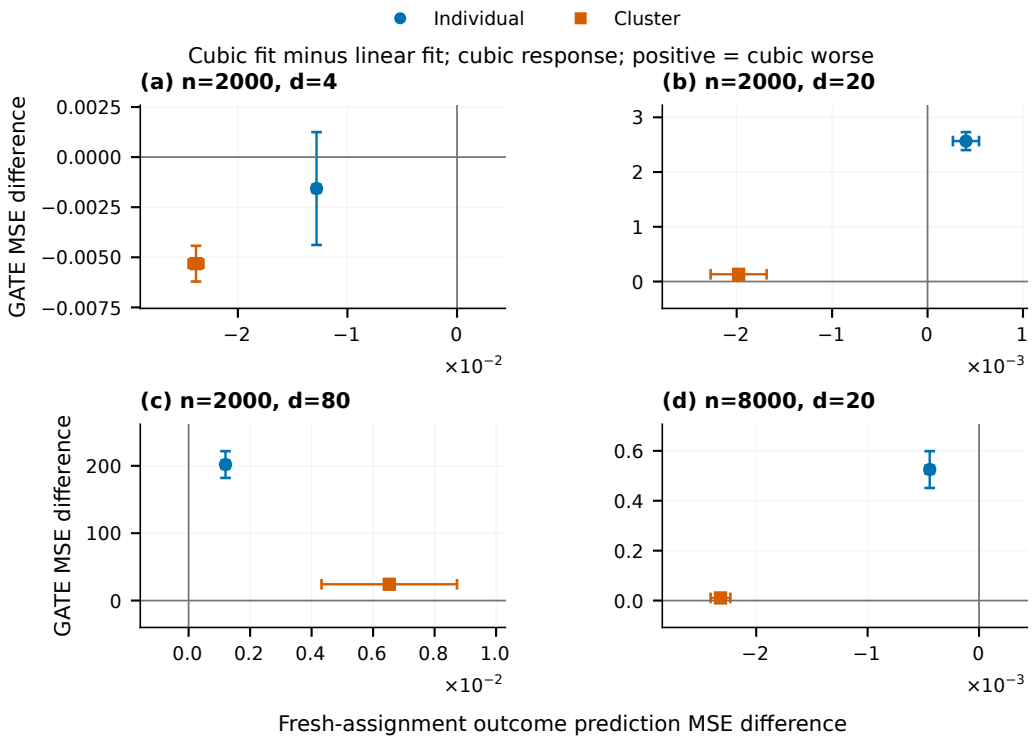


Figure 2: Better prediction can accompany worse global endpoint estimation. Regular graphs; $p = 0.5$, $\rho = 0$, $\gamma = 1$; five independent graph seeds, 400 assignment/noise attempts per graph. Points are equal-graph means and bars are supplied approximate 95% t_4 intervals across graphs, including finite Monte Carlo variation; no multiplicity adjustment. Each point compares cubic minus linear regression on the cubic response: horizontal position is the fresh independent assignment/noisy-outcome prediction MSE difference, vertical position the GATE MSE difference, both in squared outcome units. Positive means cubic is worse. Both differences use the same within-cell estimation/evaluation streams; every graph contributes 400 defined, jointly endpoint-estimable pairs. The upper-left quadrant is better prediction but worse GATE; the dense regime also exposes cases where both errors worsen. Panels use different linear scales, explicitly labeled; axis error bars are marginal intervals, not a joint confidence region. This is a finite-grid comparison, not a universal model ranking. All four planned regular size/degree regimes and both designs are included; Table 3 retains exact contrasts.

For quadratic outcomes under individual half allocation on the same regime, linear regression's signed bias is -0.002102 $[-0.02338, 0.01918]$, consistent with the complement identity in Appendix A.1. Its RMSE is nevertheless 0.2086. In the focused imbalance check on the single seed-201 graph, bias becomes -0.5755 $[-0.6016, -0.5494]$ at $p = 0.2$ and $+0.5549$ $[0.5309, 0.5789]$ at $p = 0.8$. These are conditional MC intervals. Both allocation designs on this graph and on SNAP show this sign reversal. This sensitivity check does not supply a five-graph imbalance population result or general robustness to nonlinear responses.

Interior support can hide different rollout effects (H2). With regular $n = 2000$, degree-20 graphs and individual assignment, the coupled linear/hidden-tail mechanisms yield identical training data in a mean fraction 0.697 $[0.6786, 0.7154]$ of assignments, despite a target gap of one. Linear regression's MSE is 0.04293 $[0.04027, 0.04560]$ for the linear mechanism and 1.046 $[1.003, 1.090]$ for the tail mechanism. The separate analytical union bound gives $\Pr(E) \geq 0.5975494$ and hence maximum risk at least 0.1493874 for any common, fully defined estimator across this constructed pair (Appendix A.2). At degree 80 all saved assignments satisfy E; this empirical observation does not establish probability one. Its analytical lower bound is 0.9999999795. Conditional-success Hájek risk cannot be substituted into this unconditional bound.

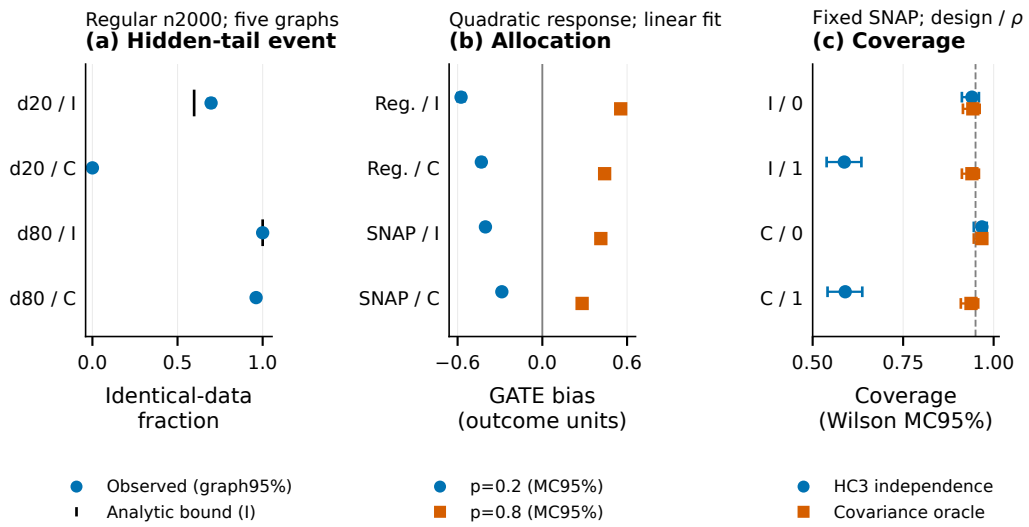


Figure 3: Support, allocation and covariance delimit regression GATE accuracy. All cells use $\gamma = 1$ and 400 assignment/noise attempts per graph. I/C denote individual/cluster assignment. (a) Linear versus hidden-tail response, regular $n = 2000$, linear fit, $p = .5$, $\rho = 0$: observed identical-data fractions with equal-graph approximate 95% t_4 intervals over five graphs (including Monte Carlo variation). Black ticks are separate analytic node-assignment event lower bounds, not estimates. A degenerate $[1, 1]$ event interval records all saved events, not probability one; per-graph Wilson bounds remain in the source. (b) Quadratic response/linear fit, $\rho = 0$: bias reverses between $p = .2/.8$ on one regular $n2000/d20$ seed201 graph and one SNAP topology. Bars are conditional normal 95% MC intervals; all 400 returns are defined/estimable. (c) Fixed SNAP, matched linear response/fit, $p = .5$: 95% interval coverage versus the nominal .95 dashed line, with Wilson MC bounds, denominator 400 per point. HC3 uses working independence; the oracle knows the simulation covariance. This is a diagnostic under the specified local noise law, not a feasible inference procedure or graph-population result. Tables 8, 10 and 11 retain numerical magnitudes and scope.

More support does not guarantee lower operational risk (H3). Clusters improve matched-linear RMSE in each $n = 2000$ regular-degree slice: cluster-minus-node differences are -0.03305, -0.04155 and -0.04964 at degrees 4, 20 and 80, with graph intervals excluding zero. These compare independent assignment/noise streams on matched graphs, not paired replicate errors. Under the degree-80 threshold response, the true-cutoff oracle's mean estimable fraction rises from 0.0060 $[0.001292, 0.01071]$ to 0.7205 $[0.7045, 0.7365]$. Operational RMSE falls from 0.9994 $[0.9958, 1.003]$ to 0.7488 $[0.7319, 0.7656]$, but finite nonestimable returns remain included and do not demonstrate identified recovery.

Flexibility can be much worse: under linear outcomes in this dense regime, the fixed spline has mean graph RMSE 820.4 $[-393.5, 2034]$ under node assignment, with no estimable contrasts, and 3946 $[3200, 4693]$ under clusters, with mean estimable fraction 0.269. The negative symmetric lower bound signals imprecision of this unstable nonnegative metric; it is retained rather than truncated. An empirical zero estimable fraction does not establish a zero underlying probability. Partition support and estimability gains therefore do not give uniform flexibility or design gains.

Correct mapping can cost more variance than it repairs bias (H4). For two-hop outcomes under node assignment, two-hop-minus-one-hop GATE MSE is -0.1165 $[-0.1213, -0.1117]$ at $n = 2000$, degree 4, but $+0.5548$ $[0.4763, 0.6334]$ at degree 20. Increasing size to $n = 8000$ at degree 20 reverses the difference again, to -0.03851 $[-0.05108, -0.02594]$. Correct features remove conditional mean bias when estimable, yet their endpoint-variance cost can dominate. A mechanism-aware fit is a specification control, not a risk lower bound. Table 4 retains every regular

mapping comparison and the threshold/spline operational risks; Table 5 covers the complete topology inventory.

Covariance changes precision and defeats working-independence intervals (H4). Under the specified unit-marginal-variance $\rho = 1$ noise, matched-linear RMSE on degree-20 regular graphs rises by 0.02123 $[0.01630, 0.02617]$ under node assignment and 0.02434 $[0.01213, 0.03654]$ under clusters. On block-model graphs, cluster-assignment HC3 coverage is 0.8395 $[0.8223, 0.8567]$, versus oracle coverage 0.9555 $[0.9475, 0.9635]$. These covariance-law results do not determine variance changes for arbitrary network errors.

On the fixed SNAP graph, matched-linear node/cluster RMSE rises from $0.1114/0.05444$ at $\rho = 0$ to $0.2579/0.1333$ at $\rho = 1$. At $\rho = 1$, HC3 coverage is 0.5875 $[0.5387, 0.6347]$ and 0.5900 $[0.5412, 0.6371]$, versus oracle 0.9400 $[0.9123, 0.9594]$ and 0.9375 $[0.9094, 0.9573]$. These are conditional MC/Wilson intervals, using 400 finite interval indicators per design. Oracle covariance is known only in simulation; its coverage does not supply feasible network inference or protect misspecified means.

Baseline controls and rare-event failures. Without interference on degree-4 regular graphs under node assignment, DIM's mean graph RMSE is 0.04978 $[0.04865, 0.05092]$, versus 0.1009 for exposure regression, 0.2613 for HT and 0.2008 for Hájek. Under linear interference, DIM's RMSE is 1.004, versus 0.1009 for matched regression. Exposure adjustment costs precision without interference but repairs this missing-spillover error. On regular $n = 2000$, degree-20 graphs at $p = 0.5$, $\rho = 0$, Hájek has no defined return in any saved node-assignment experiment; its conditional-defined RMSE is missing, not zero. Across the full grid it is undefined in 19,885 distinct assignment cells. Spline and threshold fits return finite nonestimable contrasts in 4,338 and 2,959 such cells; repeated mechanisms are not independent failure observations. No simulation cell failed or remained unrun. Table 18 gives all estimator output/event counts; Table 17 retains both no-interference and linear-response controls.

For the linear response on SNAP with $\rho = 0$, one-hop HT's exact $\log_{10}(\mathrm{RMSE})$ is 154.257 under node assignment and 12.9229 under clusters. A 400-repetition empirical error summary can miss the extreme tail and cannot certify a small true risk. These are fixed-graph, fixed-covariate, known-DGP calculations; all 306 endpoint-family values are retained, excluding two-hop from the sufficient one-hop identity (Appendix A.3). No HT/Hájek practical interval is claimed.

7 Discussion, limitations, and broader implications

Regression GATE accuracy here depends on contrast variance, response/exposure specification and estimability. Size helps in matched slices, but density can weaken endpoint information. Prediction, mechanism-aware features and cluster support each have counterexamples to guaranteed risk improvement. Support, contrast sensitivity and uncertainty therefore need joint assessment; simulation-known moments are not a practical accuracy certificate.

We evaluate fixed-feature regressions, BFS partitions and specified response/noise laws at $\gamma = 1$. Five graph draws give limited graph uncertainty; cross-family comparisons change multiple properties. SNAP is one historical topology with synthetic outcomes; imbalance uses it and one generated graph. Neither represents real platform responses. Missing edges, endogenous networks, temporal interference, changing effects, attrition, noncompliance, other treatment strengths/cluster sizes, adaptive features and GNNs are untested. Normal/delta MC intervals cannot capture missed extreme tails; graph t_4 intervals can be imprecise. Finite minimum-norm returns do not imply identification; conditional-defined risk is not unconditional algorithm risk.

Corrected evidence, all-grid integer-event checks and selected replay supersede the first run's rational-cutoff defect (Appendix C). Arithmetic replay alone had not established scientific-definition agreement. The hidden-tail obstruction and complement identity require their stated assumptions; neither is a universal nonlinear-interference result. We propose no new estimator or general inference theorem.

GATE does not measure distributed benefits or harms. Unsupported extrapolation or treating oracle uncertainty as feasible risks overconfident deployment. These findings warrant caution, not approval of an intervention. No real-platform treatment or new human-subject data are involved; public topology excludes profile/circle labels.

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A Analytical controls for interpreting the simulations

These controls specialize the supplied outcome mechanisms and fixed-feature estimators. They are explanatory calculations, not claims of new general identification or inference results. No production accuracy measurement is implied.

A.1 Quadratic response misspecification can cancel in the contrast

Consider the linear exposure model with columns $(\mathbf{1}, z, x, s)$ fitted to the quadratic mean $m(z) = 0.5x + z + \gamma s^2$. Since $P\mathbf{1} = \mathbf{1}$, complementing treatment gives $s(\mathbf{1} - z) = \mathbf{1} - s(z)$ and

$$m(\mathbf{1} - z) = m(z) + (1 + \gamma)\mathbf{1} - 2z - 2\gamma s. \tag{6}$$

Suppose the endpoint contrast is estimable. Its weights satisfy $a^\top \mathbf{1} = a^\top x = 0$ and $a^\top z = a^\top s = 1$. Complementing the assignment preserves the fitted column space and changes the sign of its endpoint weights: $a(\mathbf{1} - z) = -a(z)$. One way to see this is that $-a(z)$ satisfies the complemented contrast equations and belongs to the same column space; the minimum-norm outcome-weight solution is unique there. It follows that

$$\frac{a(z)^\top m(z) + a(\mathbf{1} - z)^\top m(\mathbf{1} - z)}{2} = 1 + \gamma = \tau_G. \tag{7}$$

Under Bernoulli$(1/2)$ assignment by nodes or fixed clusters, an assignment and its complement have equal probability. Thus signed conditional biases cancel over complement pairs for which the contrast is estimable. The estimable subset is itself complement invariant. This does not establish unbiasedness over nonestimable assignments under an arbitrary pseudoinverse convention. Nor does it make each realized bias zero or eliminate assignment variance. At $p \neq 1/2$ the complement probabilities differ, and other nonlinear response shapes need not satisfy this identity. Reporting only average signed bias could therefore hide consequential misspecification error.

A.2 A coupled response pair with different policy endpoints

Couple the linear and hidden-tail mechanisms using the same graph, covariates, treatment assignment and noise. Let

$$E = \{\max_i s_i \leq 0.85\}. \tag{8}$$

On E, the two mechanisms generate identical observed outcomes, while their targets differ by γ. For any shared estimation rule T that sees this experiment but is not told the true mechanism label, its two outputs on E coincide. For any scalar t,

$$(t - \tau_{\text{linear}})^2 + (t - \tau_{\text{tail}})^2 \geq \frac{\gamma^2}{2}. \tag{9}$$

Integrating over the coupled experiment and discarding nonnegative errors outside E yields

$$\max\{\text{MSE}_{\text{linear}}, \text{MSE}_{\text{tail}}\} \geq \frac{\gamma^2}{4} \Pr(E). \tag{10}$$

This applies to a common estimator across the constructed pair; it does not apply to an oracle given the true response mechanism. The independent prediction-evaluation experiment is not available to T. Good prediction on the interior experimental exposure distribution cannot, by itself, distinguish the two endpoint responses on E.

For a simple d-regular graph under node Bernoulli(p), each neighbor-treated count is marginally $\text{Binomial}(d, p)$. A union bound gives

$$\Pr(E) \geq \max\left\{0, 1 - n \Pr[\text{Binomial}(d, p) > 0.85d]\right\}. \tag{11}$$

This uses no independence assumption between units' exposures. It is a graph-conditional bound and is not the marginal law for cluster assignment. A zero lower bound is uninformative, not evidence that E is impossible.

A.3 Exact full-exposure HT risk under a sufficient one-hop mapping

Let N_i be the set of independent assignment units in the closed neighborhood of node i, and $k_{ij} = |N_i \cap N_j|$. These units are nodes under individual assignment and fixed partition pieces under cluster assignment. Write $m_i^a = m_i(a\mathbf{1})$ and $Q_{ij}^{ab} = m_i^a m_j^b + \Sigma_{ij}$ for $a, b \in \{0, 1\}$. Assume the one-hop exposure event reveals the corresponding endpoint mean and the additive mean-zero noise is independent of assignment, with common covariance Σ. Then HT is unbiased, so its MSE equals its variance:

$$\mathrm{Var}(\widehat{\tau}_{\mathrm{HT}}) = \frac{1}{n^2} \sum_{i,j} \big[Q_{ij}^{11}(p^{-k_{ij}} - 1) + Q_{ij}^{00}\{(1-p)^{-k_{ij}} - 1\} + (Q_{ij}^{10} + Q_{ij}^{01})\mathbf{1}\{k_{ij} > 0\} \big]. \tag{12}$$

For same-arm exposure indicators, their joint-to-marginal probability ratio is $p^{-k_{ij}}$ or $(1-p)^{-k_{ij}}$. Opposite-arm events are incompatible when the assignment-unit sets overlap and independent otherwise. Subtracting the endpoint-product contribution to the squared target yields Equation (12). In the full ordered double sum, the two cross terms can equivalently be written $2Q_{ij}^{10}$ by exchanging i, j and using covariance symmetry. This is the form evaluated blockwise by the analysis implementation.

The formula includes outcome-noise covariance as well as randomization variation. One-hop mechanisms with the same endpoint means and covariance share this risk even when their interior responses differ. The two-hop mechanism does not meet the stipulated exposure sufficiency and is excluded from this unbiased-risk calculation. Logarithmic risk output is needed when rare-event contributions exceed ordinary floating-point range; a finite Monte Carlo sample without such events cannot certify a small HT risk.

B Additional response and fitting details

The normalized sigmoid response is

$$h_i = \frac{\mathrm{expit}(12(s_i - 0.5)) - \mathrm{expit}(-6)}{\mathrm{expit}(6) - \mathrm{expit}(-6)}.$$

All non-DIM regressions contain an intercept, own treatment and the fixed covariate. The quadratic/cubic models include all exposure powers through their degree. The two-hop model adds s and t. The spline adds a cubic B-spline basis with five equally spaced knots from zero to one, no bias basis column and linear extrapolation; its basis and knots are fixed before outcomes. The threshold oracle adds the true $\mathbf{1}\{s \geq 0.75\}$ feature. Singular values are retained above machine precision times the larger design-matrix dimension times its largest singular value. Endpoint contrasts use the same feature pipeline at both global assignments. HC3 clips $1 - h_{ii}$ below 10^{-10} for numerical stability; this does not repair misspecification or correlated errors. Complete estimator/source definitions and denominator policies are supplied in `analysis/methods.md` and `code/src/gatebench/core.py`.

C Graph construction, correction, and execution verification

Graph construction and fixed inputs. Regular graphs use the declared size/degree. Erdős–Rényi edges have probability $20/(n-1)$. Barabási–Albert graphs attach ten edges per new node, giving realized mean degree near 20. The block model has eight equal blocks, expected degree 20 and mixing $\mu = 0.15$: with block size $b = n/8$, within/between probabilities are $20(1-\mu)/(b-1)$ and $20\mu/(n-b)$. Table 1 reports all 36 measured graphs/partitions; these structural attributes match all 102 corrected graph/config metadata records in `analysis/v2_dependency_audit.json`. For graph seed g, fixed covariates use seed $g+123456$ and BFS partitioning uses $g+97$. Experimental assignment/noise/prediction seeds and master-namespaced pairing identifiers are stored per replicate. Models/mechanisms share streams within a cell; designs/panels use independent replicate streams.

Scientific-definition correction. The first full execution was superseded after a final scientific check found that normalized sparse accumulation could put $17/20$ above 0.85 and $60/80$ below 0.75. This could misclassify hidden-tail support and threshold outcomes/features despite successful

Table 1: Measured graph and fixed partition inventory. Each row is one processed graph used by both individual and cluster assignment in the corrected core execution; n and m count vertices and undirected edges, $\bar{d}$ is mean degree, K counts assignment clusters and f_{in} is the fraction of edges within those clusters. All listed graphs have zero isolates. Cluster size 32 is the partition construction target, not n/K. Synthetic regimes each have five independent graph seeds; SNAP Facebook is one fixed public topology with simulated outcomes, and its seed labels covariate/partition generation. Each graph/design core cell used 400 assignment/noise replicates, $p = 0.5$, $\rho = 0$, $\gamma = 1$ and eight response mechanisms. No across-graph averaging or uncertainty estimate is shown. All 102 corrected graph/config metadata records match the preserved original structural content; this table reports the 72 core cells. It supplies structural scope, not an estimator-accuracy conclusion.

Graph	Seed	n	m	$\bar{d}$	K	f_{in}	Isolates
Regular	201	2,000	4,000	4.0	145	0.470	0
Regular	202	2,000	4,000	4.0	149	0.470	0
Regular	203	2,000	4,000	4.0	146	0.469	0
Regular	204	2,000	4,000	4.0	140	0.469	0
Regular	205	2,000	4,000	4.0	149	0.468	0
Regular	201	2,000	20,000	20.0	97	0.107	0
Regular	202	2,000	20,000	20.0	97	0.107	0
Regular	203	2,000	20,000	20.0	111	0.107	0
Regular	204	2,000	20,000	20.0	111	0.107	0
Regular	205	2,000	20,000	20.0	98	0.106	0
Regular	201	2,000	80,000	80.0	75	0.038	0
Regular	202	2,000	80,000	80.0	77	0.038	0
Regular	203	2,000	80,000	80.0	75	0.038	0
Regular	204	2,000	80,000	80.0	71	0.038	0
Regular	205	2,000	80,000	80.0	72	0.038	0
Regular	201	8,000	80,000	20.0	410	0.098	0
Regular	202	8,000	80,000	20.0	398	0.098	0
Regular	203	8,000	80,000	20.0	409	0.098	0
Regular	204	8,000	80,000	20.0	396	0.098	0
Regular	205	8,000	80,000	20.0	396	0.098	0
Erdős–Rényi	201	2,000	20,316	20.3	115	0.106	0
Erdős–Rényi	202	2,000	19,703	19.7	131	0.107	0
Erdős–Rényi	203	2,000	20,028	20.0	111	0.107	0
Erdős–Rényi	204	2,000	20,005	20.0	117	0.106	0
Erdős–Rényi	205	2,000	19,909	19.9	111	0.107	0
SBM	201	2,000	20,135	20.1	107	0.148	0
SBM	202	2,000	20,224	20.2	110	0.146	0
SBM	203	2,000	19,953	20.0	108	0.144	0
SBM	204	2,000	19,833	19.8	108	0.147	0
SBM	205	2,000	20,088	20.1	94	0.151	0
Barabási–Albert	201	2,000	19,900	19.9	172	0.105	0
Barabási–Albert	202	2,000	19,900	19.9	178	0.105	0
Barabási–Albert	203	2,000	19,900	19.9	178	0.103	0
Barabási–Albert	204	2,000	19,900	19.9	178	0.104	0
Barabási–Albert	205	2,000	19,900	19.9	169	0.104	0
SNAP Facebook	201	4,039	88,234	43.7	430	0.217	0

arithmetic audits and selected replay. The corrected one-hop computation counts treated neighbors before dividing by degree, with isolates still using own treatment. It preserves graphs, configurations, seeds, 400 repetitions and all 102 cells; every empirical and exact-risk artifact was regenerated under frozen corrected source. An independent integer-event checker replayed all 40,800 saved assignments using $4k \geq 3d$ for the threshold and $20k > 17d$ for hidden-tail activation, checking stored maxima/support counts without regression refitting. Its receipt is `analysis/boundary_audit.json`. Original outputs/source/receipts remain superseded history in `analysis/roundoff_v1/`; pilot boundaries were not repaired and pilots supply calibration only.

Audits and selected clean replay. All 102 cells and 306 exact-risk records passed completeness/provenance checks. The internal raw-to-summary audit checked 73,440 scalar and 48,960 paired moments, graph uncertainty and coverage/failure denominators; 280 all-undefined graph/model groups retain missing bounds. Supplementary checks cover 676,164 comparisons and 5,256 independent-stream contrasts. The 13 scientific tests include endpoint targets, conditional error algebra, complement/coupling controls, exact HT enumeration, rational cutoff equality and failure-aware scoring. These are internal validity checks, not independent research review.

The corrected clean replay reran four predeclared cells: regular $n = 2000$, degree 4 or 80, seed 201, both assignment designs, all 400 repetitions. Every column and original/replay Parquet hash matched for 115,200 outputs; measured elapsed time was 84.09 seconds. Source/hash receipts are `analysis/reproduction_audit.json`. This checks deterministic selected-cell replay, not an independent implementation or full-study reproduction. Source/config/raw/aggregate hashes and the complete failure ledger are preserved separately.

Measured compute and its limits. Corrected CPU execution used three disjoint 34-cell shards with one BLAS/OpenMP/MKL thread. Resource creation-to-completion lifecycle was 923.21 seconds; summed reported running time was 2427.93 seconds and reported CPU spend 2590.74 seconds. These definitions differ from per-function sums and include/exclude different overheads; they must not be conflated into one wall-time estimate. Maximum recorded simulation/exact-risk process RSS was 496,880 KiB; the later summary/audit stage recorded cumulative child-process peak RSS 522,460 KiB. Neither is a container-wide peak measurement. Per-panel wall/CPU/RSS/bytes, corrected finalization/replay and superseded resource records are `analysis/final_execution.json`, `analysis/execution_reconciliation.md` and `analysis/resource_jobs.json`; preliminary calibration timings are recorded separately. The original 7.30-hour allowance was a planning estimate. There is no instrumented total for every exploratory sandbox action or language-model computation. Root reproduction commands use the pinned environment and clean workspace; timing need not match on another host.

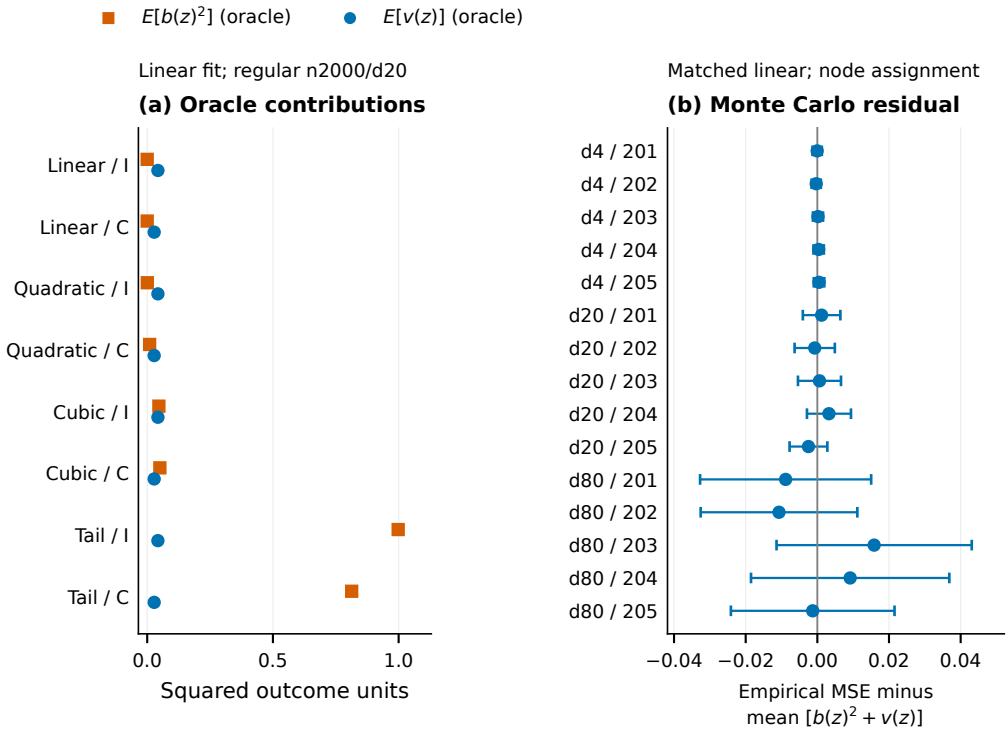


Figure 4: Conditional error decomposition and finite-simulation agreement. All cells use $\gamma = 1$ and 400 assignment/noise attempts per graph. I/C denote individual/cluster assignment. (a) Linear fit on four regular $n = 2000, d = 20$ response mechanisms, $p = .5$, $\rho = 0$. Points are the supplied equal-graph means of $E[b(z)^2]$ and $E[v(z)]$ in squared outcome units; bars are approximate 95% t_4 graph intervals including Monte Carlo variation. $b(z)$ and $v(z)$ are simulation-known conditional bias and noise variance. Squared signed bias is a different quantity. All selected outputs are defined and endpoint-estimable. (b) All 15 planned regular $n = 2000$, degree4/20/80 graph draws, individual assignment, matched linear response/fit. Each point is empirical MSE minus the sample mean of $b(z)^2 + v(z)$, with its supplied pointwise normal MC95% interval. The residual need not be exactly zero at R400. The groups are dependent across response/model comparisons; no independence test, multiplicity-adjusted rejection or empirical proof of the algebraic identity is claimed. Tables 13 and 14 retain the component magnitudes and full-study diagnostic scope.

D Numerical companions and complete diagnostic displays

These displays retain the corrected source values, conditioning and uncertainty behind Sections 6 and the analytical diagnostics. Figure/table generation routes and full selected source rows are in `figures/display-map.json`; the full-grid outputs remain in `results/summary/` and `results/diagnostics/`. Figure 3 summarizes the support, allocation and covariance limits; Figure 4 separates squared conditional bias from noise variance. Tables 4, 18 and 20 retain adverse, unsupported and extreme-risk cases.

Table 2: Numerical companion to Figure 1. Regular graphs, five seeds and 400 assignment/noise attempts per graph, $p = 0.5$, $\rho = 0$, $\gamma = 1$. Bias and RMSE are equally weighted means of graph summaries in outcome units; brackets are approximate 95% t_4 graph intervals. $\sqrt{E[v]}$ is the square root after averaging the simulation-known conditional noise variance, distinct from mean graph RMSE; it has no interval here. $E[b^2]$ is mean squared assignment-conditional bias, in squared outcome units, distinct from squared signed bias. All returns here are defined and endpoint-estimable. I/C mean individual/cluster assignment. The second block fixes $n = 2000, d = 20$ and I assignment and uses the linear fitted model on each named response (GATE 2 for linear/quadratic/cubic, 3 for hidden tail); near-zero quadratic signed bias does not imply absence of conditional bias. All values are rounded to three significant digits.

Setting	Signed bias [95%]	RMSE [95%]	$\sqrt{E[v]}$	$E[b^2]$
Linear response, matched linear fit				
2000/4/I	0.000443 [-0.00377, 0.00465]	0.101 [0.0989, 0.103]	0.1	1.5e-30
2000/4/C	0.000184 [-0.00428, 0.00465]	0.0678 [0.0646, 0.071]	0.0686	6.2e-30
2000/20/I	-0.00259 [-0.024, 0.0188]	0.207 [0.201, 0.214]	0.206	4e-30
2000/20/C	0.00354 [-0.0115, 0.0185]	0.166 [0.159, 0.172]	0.167	2.9e-30
2000/80/I	0.0166 [-0.0111, 0.0442]	0.412 [0.395, 0.43]	0.411	2.1e-30
2000/80/C	0.0125 [-0.00615, 0.0311]	0.362 [0.345, 0.38]	0.363	1.9e-30
8000/20/I	0.00306 [-0.00409, 0.0102]	0.101 [0.0978, 0.104]	0.103	7.6e-30
8000/20/C	-0.00171 [-0.00749, 0.00408]	0.0832 [0.0801, 0.0863]	0.0825	3.4e-30
Linear fit, I assignment; response varied at n=2000, d=20				
Linear	-0.00259 [-0.024, 0.0188]	0.207 [0.201, 0.214]	0.206	3.98e-30
Quadratic	-0.0021 [-0.0234, 0.0192]	0.209 [0.203, 0.214]	0.206	0.000476
Cubic	-0.216 [-0.237, -0.194]	0.301 [0.284, 0.318]	0.206	0.0465
Hidden tail	-1 [-1.02, -0.98]	1.02 [1, 1.04]	0.206	0.998

Table 3: Paired cubic-versus-linear prediction and GATE comparisons underlying Figure 2. Cubic response on regular graphs; five graph seeds, 400 defined, jointly estimable within-stream pairs per graph, $p = 0.5$, $\rho = 0$, $\gamma = 1$. Differences are cubic minus linear in squared outcome units; positive is worse for cubic. Prediction evaluates a fresh independent assignment and noisy outcome on the same fixed graph/covariates. Values are equal-graph mean paired differences; brackets are approximate 95% t_4 intervals across the five graph summaries. They are not individual-experiment GATE intervals or multiplicity-adjusted tests. All four planned regular regimes and both assignments are shown. I/C denote individual/cluster assignment. Each figure panel has its own linear scales; this table retains numerical magnitudes.

n/d/Design	Prediction MSE difference [95%]	GATE MSE difference [95%]
2000/4/I	-0.01282 [-0.01317, -0.01247]	-0.001569 [-0.004385, 0.001247]
2000/4/C	-0.02382 [-0.0245, -0.02315]	-0.005317 [-0.006208, -0.004426]
2000/20/I	0.0004027 [0.0002658, 0.0005395]	2.566 [2.402, 2.73]
2000/20/C	-0.00198 [-0.002273, -0.001686]	0.1324 [0.1054, 0.1594]
2000/80/I	0.0012 [0.001074, 0.001326]	201.9 [182.1, 221.8]
2000/80/C	0.006526 [0.004323, 0.008729]	24.17 [22.16, 26.19]
8000/20/I	-0.0004426 [-0.0004799, -0.0004054]	0.5251 [0.4513, 0.5988]
8000/20/C	-0.002321 [-0.002409, -0.002232]	0.01049 [0.003558, 0.01742]

Table 4: Conditional value and failure of response/mapping-aware fits on regular graphs, $p = 0.5$, $\rho = 0$, $\gamma = 1$, 400 assignment/noise attempts per graph, five graph seeds per row. Brackets are supplied approximate 95% t_4 intervals across equally weighted graph summaries. I/C denote individual/cluster assignment. Panel A is the within-stream paired GATE MSE difference (two-hop fit minus linear fit) on the two-hop response; positive means the mapping-aware fit is worse. Every graph has 400 defined, jointly estimable pairs. Panels B/C report the mean graph operational RMSE (outcome units), retaining all finite minimum-norm returns, and the mean endpoint-estimable fraction f_E; all 400 returns per graph are finite. B uses the known threshold feature on the threshold response; C uses a fixed spline on the linear response. All risks are operational, not estimable-only. Untruncated symmetric intervals can extend below zero for unstable nonnegative RMSE; this signals low precision. Differences are in squared outcome units. All four planned regular size/degree regimes are retained in A; B/C vary degree at $n = 2000$.

Mapping, support and flexible-fit boundaries				
A: Two-hop response; two-hop fit minus linear fit				
n	d	Design	GATE MSE difference [95%]	
2000	4	I	-0.116 [-0.121, -0.112]	
2000	4	C	-0.0355 [-0.0376, -0.0333]	
2000	20	I	0.555 [0.476, 0.633]	
2000	20	C	0.395 [0.363, 0.426]	
2000	80	I	14.8 [13.5, 16.1]	
2000	80	C	10.9 [9.78, 12]	
8000	20	I	-0.0385 [-0.0511, -0.0259]	
8000	20	C	-0.0523 [-0.0719, -0.0327]	
B: Threshold response; threshold oracle				
n	d	Design	Mean RMSE [95%]	f_E
2000	4	I	0.0659 [0.0643, 0.0676]	1
2000	4	C	0.0502 [0.0469, 0.0536]	1
2000	20	I	0.169 [0.162, 0.175]	1
2000	20	C	0.123 [0.113, 0.133]	1
2000	80	I	0.999 [0.996, 1]	0.006
2000	80	C	0.749 [0.732, 0.766]	0.7205
C: Linear response; fixed spline				
n	d	Design	Mean RMSE [95%]	f_E
2000	4	I	0.136 [0.129, 0.143]	1
2000	4	C	0.0796 [0.0748, 0.0845]	1
2000	20	I	34.5 [30.2, 38.9]	0.9995
2000	20	C	1.02 [0.504, 1.53]	1
2000	80	I	820 [-394, 2.03e+03]	0
2000	80	C	3.95e+03 [3.2e+03, 4.69e+03]	0.269

Table 5: Regression accuracy across the complete core topology inventory, $p = 0.5$, $\rho = 0$, $\gamma = 1$, 400 assignment/noise attempts per graph. Entries are GATE RMSE in outcome units with supplied 95% uncertainty bounds. Synthetic entries are equal-graph mean RMSEs with approximate t_4 intervals across five graph seeds. SNAP is one fixed full graph: its entries are graph-conditional RMSEs with delta-method Monte Carlo intervals, not graph-population intervals. All 400 outputs per graph in these selected mechanism/model pairs are finite and endpoint-estimable. The column headings identify response/fitted model: linear/linear is matched one-hop, cubic/linear is misspecified response shape, and two-hop/two-hop uses the correct exposure mapping. These columns change both response and fit; they do not isolate a model-comparison effect. Reg., ER, SBM and BA mean regular, Erdos–Renyi, stochastic-block and Barabasi–Albert graphs; 2k/8k are 2000/8000 nodes. I/C denote individual/cluster assignment. Nonregular families have target degree 20; topology comparisons change multiple properties and are descriptive, not isolated degree effects. Graph intervals include finite Monte Carlo variation. Fixed covariates/partitions and synthetic outcomes limit generalization. Full graph and partition properties are in Table 1.

Topology / design	Linear / linear	Cubic / linear	Two-hop / two-hop
Reg. 2k d4 / I	0.101 [0.0989, 0.103]	0.141 [0.137, 0.146]	0.185 [0.18, 0.191]
Reg. 2k d4 / C	0.0678 [0.0646, 0.071]	0.108 [0.102, 0.114]	0.115 [0.113, 0.116]
Reg. 2k d20 / I	0.207 [0.201, 0.214]	0.301 [0.284, 0.318]	0.909 [0.867, 0.95]
Reg. 2k d20 / C	0.166 [0.159, 0.172]	0.277 [0.266, 0.289]	0.782 [0.766, 0.798]
Reg. 2k d80 / I	0.412 [0.395, 0.43]	0.47 [0.453, 0.488]	3.9 [3.72, 4.07]
Reg. 2k d80 / C	0.362 [0.345, 0.38]	0.45 [0.44, 0.461]	3.35 [3.19, 3.52]
Reg. 8k d20 / I	0.101 [0.0978, 0.104]	0.234 [0.226, 0.241]	0.441 [0.434, 0.449]
Reg. 8k d20 / C	0.0832 [0.0801, 0.0863]	0.207 [0.202, 0.212]	0.385 [0.366, 0.405]
ER 2k / I	0.196 [0.189, 0.203]	0.287 [0.279, 0.296]	0.895 [0.848, 0.942]
ER 2k / C	0.166 [0.157, 0.174]	0.278 [0.271, 0.284]	0.753 [0.733, 0.773]
SBM 2k / I	0.199 [0.193, 0.205]	0.288 [0.279, 0.298]	0.706 [0.694, 0.718]
SBM 2k / C	0.134 [0.127, 0.141]	0.254 [0.249, 0.259]	0.27 [0.258, 0.282]
BA 2k / I	0.176 [0.173, 0.179]	0.271 [0.263, 0.28]	0.798 [0.771, 0.826]
BA 2k / C	0.147 [0.141, 0.153]	0.279 [0.262, 0.295]	0.693 [0.661, 0.724]
SNAP (fixed) / I	0.111 [0.104, 0.119]	0.211 [0.198, 0.224]	0.241 [0.224, 0.258]
SNAP (fixed) / C	0.0544 [0.051, 0.0578]	0.134 [0.124, 0.143]	0.0741 [0.0694, 0.0788]

Table 6: Operational empirical performance on the full fixed SNAP ego-Facebook graph ($n = 4039$), linear response, 400 assignment/noise attempts per row, $p = 0.5$, $\rho = 0$, $\gamma = 1$. Bias, error SD and finite-run RMSE use outcome units. Bias/RMSE brackets are approximate conditional 95% Monte Carlo bounds (normal bias, delta-method RMSE), not graph-population intervals. D/E are exact defined/valid return counts out of 400 (valid means endpoint-estimable for regressions; for HT/Hájek it equals defined); unsupported finite minimum-norm returns remain in operational error. Exposure Hájek errors condition on a defined ratio; undefined returns are excluded and counted, never set to zero. An absent error means all returns are undefined. HT values are finite-run empirical errors, not reliable rare-event risk estimates: the exact one-hop HT risks are separately reported in Table 7. The threshold oracle knows a threshold feature but is misspecified for this linear response. No universal model ranking or real-platform effect validation follows. Values are rounded to three significant digits; SD describes replicate dispersion. No Monte Carlo error bounds are shown for HT/Hájek because rare exposure tails can invalidate that interpretation.

Model	Bias [MC95%]	Error SD	Empirical RMSE [MC95%]	D/E
Individual assignment; linear response				
Mean difference	-0.999 [-1, -0.996]	0.0388	1 [0.996, 1]	400/400
Linear OLS	-0.006 [-0.0169, 0.00491]	0.111	0.111 [0.104, 0.119]	400/400
Quadratic OLS	-0.00482 [-0.0164, 0.00676]	0.118	0.118 [0.109, 0.127]	400/400
Cubic OLS	-0.00441 [-0.0204, 0.0115]	0.163	0.163 [0.151, 0.174]	400/400
Fixed spline	-0.0103 [-0.0267, 0.00604]	0.167	0.167 [0.156, 0.179]	400/400
Two-hop OLS	-0.0126 [-0.0363, 0.011]	0.241	0.241 [0.224, 0.258]	400/400
Threshold oracle	-0.602 [-0.61, -0.593]	0.0852	0.608 [0.6, 0.616]	400/400
Exposure HT	-1.21	4.69	4.84	400/400
Exposure Hájek	0.0056	0.666	0.665	400/400
Cluster assignment; linear response				
Mean difference	-0.689 [-0.693, -0.684]	0.0438	0.69 [0.686, 0.694]	400/400
Linear OLS	0.00596 [0.000649, 0.0113]	0.0542	0.0544 [0.051, 0.0578]	400/400
Quadratic OLS	0.00672 [0.00123, 0.0122]	0.056	0.0563 [0.0529, 0.0598]	400/400
Cubic OLS	0.00245 [-0.00597, 0.0109]	0.086	0.0859 [0.0802, 0.0915]	400/400
Fixed spline	0.00189 [-0.00739, 0.0112]	0.0947	0.0946 [0.0882, 0.101]	400/400
Two-hop OLS	0.00472 [-0.00254, 0.012]	0.074	0.0741 [0.0694, 0.0788]	400/400
Threshold oracle	-0.415 [-0.42, -0.41]	0.0541	0.419 [0.413, 0.424]	400/400
Exposure HT	-0.264	4.02	4.02	400/400
Exposure Hájek	0.0122	0.276	0.276	399/399

Table 7: Finite simulation can miss full-exposure HT rare-event risk. Full fixed SNAP graph, linear response, $p = 0.5$, $\rho = 0$, $\gamma = 1$. The empirical column is RMSE over 400 assignment/noise draws; the exact column integrates assignment/noise conditional on the same graph/covariates. An exact entry k represents RMSE 10^k in outcome units. No Monte Carlo interval is claimed for exact risk; a finite-run empirical error bar cannot capture unobserved rare-event risk. This one-hop unbiased-risk identity excludes the two-hop response. All exact core graph/design risks are retained in Table 19.

Assignment	Finite R400 RMSE	Exact $\log_{10}$(RMSE)
Individual	4.837	154.25653
Cluster	4.023	12.92295

Table 8: Hidden-tail support and its analytic lower bound, regular graphs, $p = .5$, $\rho = 0$, linear fit. All cells use $\gamma = 1$ and 400 assignment/noise attempts per graph. I/C denote individual/cluster assignment. Five independent graph draws per synthetic regime; entries are equally weighted means of graph summaries with approximate 95% t_4 intervals, including finite Monte Carlo variation. The event is no exposure strictly above .85; on it linear and hidden-tail responses share the observed data, while their endpoint targets differ by one. The supplied rule is fully defined on all 400 attempts per graph and returns the same estimate on that event. The coupled MSEs are in Table 9. The analytic column is a node-assignment union-bound lower bound for the event, not a confidence bound or measured frequency; a bound q implies max of the two unconditional MSEs $\geq q/4$. No such analytic bound is asserted here for clusters. An empirical graph interval $[1, 1]$ describes all saved events, not probability one; per-graph Wilson bounds are retained in the source ledger. No bound is applied to conditional-success Hájek risk.

n/d/Design	Event fraction [graph95%]	Analytic event lower bound
2000/4/I	0 [0, 0]	0.0000000000
2000/4/C	0 [0, 0]	—
2000/20/I	0.697 [0.679, 0.715]	0.5975494385
2000/20/C	0 [0, 0]	—
2000/80/I	1 [1, 1]	0.9999999795
2000/80/C	0.962 [0.954, 0.969]	—
8000/20/I	0.198 [0.175, 0.221]	0.0000000000
8000/20/C	0 [0, 0]	—

Table 9: Coupled linear/hidden-tail operational MSEs for the event/support inventory in Table 8. All cells use $\gamma = 1$ and 400 assignment/noise attempts per graph. I/C denote individual/cluster assignment. Five independent graph draws per synthetic regime; entries are equally weighted means of graph summaries with approximate 95% t_4 intervals, including finite Monte Carlo variation. All pairs use the same assignment/noise within each graph and both estimates are fully defined. The fitted rule is linear, $p = .5$, $\rho = 0$, and the targets differ by one. Units are squared outcome units. These are supplied graph means and intervals, not uncertainty bounds on the separate analytic lower bound. No conditional-success Hájek risk is used.

n/d/Design	Linear MSE [graph95%]	Hidden-tail MSE [graph95%]
2000/4/I	0.0102 [0.00978, 0.0106]	0.262 [0.255, 0.269]
2000/4/C	0.0046 [0.00417, 0.00504]	0.0699 [0.0663, 0.0736]
2000/20/I	0.0429 [0.0403, 0.0456]	1.05 [1, 1.09]
2000/20/C	0.0274 [0.0253, 0.0296]	0.835 [0.798, 0.871]
2000/80/I	0.17 [0.156, 0.184]	1.14 [1.08, 1.19]
2000/80/C	0.132 [0.119, 0.144]	1.11 [1.07, 1.14]
8000/20/I	0.0102 [0.00955, 0.0108]	1 [0.987, 1.02]
8000/20/C	0.00693 [0.00641, 0.00744]	0.84 [0.825, 0.855]

Table 10: Allocation reverses signed misspecification bias on the focused imbalance slice: quadratic response, linear fit, $\rho = 0$. All cells use $\gamma = 1$ and 400 assignment/noise attempts per graph. I/C denote individual/cluster assignment. SNAP and the seed201 imbalance graph are fixed: their bounds are conditional Monte Carlo intervals, not graph-population intervals. All 400 outputs per graph are defined and endpoint-estimable. Bias and RMSE have outcome units; brackets are supplied normal/delta-method 95% Monte Carlo bounds. Independent assignment/noise streams are used in the imbalance panel. There is one regular graph seed, not five; these are conditional two-graph results.

Fixed graph / design	p	Signed bias [MC95%]	RMSE [MC95%]
Regular 2k d20 / I	0.2	-0.576 [-0.602, -0.549]	0.634 [0.609, 0.659]
Regular 2k d20 / I	0.8	0.555 [0.531, 0.579]	0.606 [0.583, 0.63]
Regular 2k d20 / C	0.2	-0.432 [-0.454, -0.409]	0.489 [0.466, 0.511]
Regular 2k d20 / C	0.8	0.441 [0.419, 0.464]	0.497 [0.476, 0.519]
SNAP / I	0.2	-0.403 [-0.42, -0.386]	0.438 [0.421, 0.455]
SNAP / I	0.8	0.414 [0.396, 0.432]	0.454 [0.435, 0.473]
SNAP / C	0.2	-0.287 [-0.297, -0.276]	0.306 [0.295, 0.316]
SNAP / C	0.8	0.282 [0.272, 0.292]	0.299 [0.289, 0.31]

Table 11: Local correlated noise changes error and interval coverage, matched linear response/fit, $p = .5$. All cells use $\gamma = 1$ and 400 assignment/noise attempts per graph. I/C denote individual/cluster assignment. Five independent graph draws per synthetic regime; entries are equally weighted means of graph summaries with approximate 95% t_4 intervals, including finite Monte Carlo variation. SNAP is one fixed topology: its bounds are conditional Monte Carlo intervals, not graph-population intervals. All 400 outputs and both intervals per graph are finite and endpoint-estimable (each coverage denominator is 400). Entries are RMSE in outcome units and coverage fractions; SNAP coverage uses Wilson 95% bounds and RMSE uses delta-method bounds. The $\rho = 1$ row-normalized local noise law preserves unit marginal variances. HC3 assumes working independence; the oracle uses the true simulation covariance. Near-nominal oracle coverage is a diagnostic, not a feasible network inference method. No arbitrary-covariance guarantee follows.

Graph / design / ρ	RMSE [95%]	HC3 coverage [95%]	Oracle coverage [95%]
Regular 2k d20 / I / 0	0.207 [0.201, 0.214]	0.955 [0.942, 0.968]	0.954 [0.941, 0.967]
Regular 2k d20 / I / 1	0.228 [0.22, 0.237]	0.92 [0.915, 0.926]	0.948 [0.935, 0.96]
Regular 2k d20 / C / 0	0.166 [0.159, 0.172]	0.949 [0.941, 0.957]	0.944 [0.933, 0.955]
Regular 2k d20 / C / 1	0.19 [0.184, 0.196]	0.919 [0.907, 0.931]	0.947 [0.934, 0.96]
SBM 2k / I / 0	0.199 [0.193, 0.205]	0.956 [0.944, 0.967]	0.955 [0.944, 0.966]
SBM 2k / I / 1	0.227 [0.217, 0.237]	0.91 [0.889, 0.932]	0.948 [0.931, 0.964]
SBM 2k / C / 0	0.134 [0.127, 0.141]	0.943 [0.934, 0.952]	0.944 [0.93, 0.958]
SBM 2k / C / 1	0.187 [0.18, 0.194]	0.839 [0.822, 0.857]	0.956 [0.947, 0.964]
SNAP (fixed) / I / 0	0.111 [0.104, 0.119]	0.94 [0.912, 0.959]	0.943 [0.915, 0.961]
SNAP (fixed) / I / 1	0.258 [0.24, 0.276]	0.588 [0.539, 0.635]	0.94 [0.912, 0.959]
SNAP (fixed) / C / 0	0.0544 [0.051, 0.0578]	0.968 [0.945, 0.981]	0.968 [0.945, 0.981]
SNAP (fixed) / C / 1	0.133 [0.124, 0.142]	0.59 [0.541, 0.637]	0.938 [0.909, 0.957]

Table 12: Mean 95% interval widths for Table 11, in outcome units. All cells use $\gamma = 1$ and 400 assignment/noise attempts per graph. I/C denote individual/cluster assignment. The rows share its matched-model scope and exact 400-per-graph finite/estimable denominators. Width is $2(1.96)$ times the supplied mean SE: a deterministic linear unit conversion, with no new uncertainty estimate or averaging. Synthetic entries are equal-graph means; SNAP is conditional on its one topology. HC3 is working independence; the known-covariance oracle is diagnostic only.

Graph / design / ρ	Mean HC3 width	Mean oracle width
Regular 2k d20 / I / 0	0.8098	0.8087
Regular 2k d20 / I / 1	0.8091	0.8932
Regular 2k d20 / C / 0	0.6541	0.6527
Regular 2k d20 / C / 1	0.6528	0.7305
SBM 2k / I / 0	0.7918	0.7906
SBM 2k / I / 1	0.7917	0.9071
SBM 2k / C / 0	0.5286	0.5282
SBM 2k / C / 1	0.5282	0.747
SNAP (fixed) / I / 0	0.4317	0.4312
SNAP (fixed) / I / 1	0.4274	0.9652
SNAP (fixed) / C / 0	0.2242	0.2239
SNAP (fixed) / C / 1	0.2228	0.4765

Table 13: Oracle conditional MSE contributions for the linear fitted model on four responses, regular $n = 2000, d = 20$, $p = .5$, $\rho = 0$. All cells use $\gamma = 1$ and 400 assignment/noise attempts per graph. I/C denote individual/cluster assignment. Five independent graph draws per synthetic regime; entries are equally weighted means of graph summaries with approximate 95% t_4 intervals, including finite Monte Carlo variation. $b(z)$ is the assignment-conditional mean bias and $v(z)$ the known conditional noise variance; $E[b(z)^2]$ is not squared signed bias. All selected returns are finite and endpoint-estimable. The assignment expectation of $b(z)^2 + v(z)$, estimated here by its supplied sample moments, is the noise-integrated MSE diagnostic. Units are squared outcome units. These simulation-known quantities are not observable certificates of robust endpoint extrapolation.

Response / design	$E[b(z)^2]$ [graph95%]	$E[v(z)]$ [graph95%]
Linear / I	3.98e-30 [2.43e-30, 5.52e-30]	0.0426 [0.0424, 0.0427]
Linear / C	2.9e-30 [1.9e-30, 3.91e-30]	0.0277 [0.0275, 0.0279]
Quadratic / I	0.000476 [0.000409, 0.000544]	0.0426 [0.0424, 0.0427]
Quadratic / C	0.00966 [0.00892, 0.0104]	0.0277 [0.0275, 0.0279]
Cubic / I	0.0465 [0.0455, 0.0474]	0.0426 [0.0424, 0.0427]
Cubic / C	0.0504 [0.0468, 0.054]	0.0277 [0.0275, 0.0279]
Hidden tail / I	0.998 [0.998, 0.998]	0.0426 [0.0424, 0.0427]
Hidden tail / C	0.813 [0.796, 0.831]	0.0277 [0.0275, 0.0279]

Table 14: Finite-Monte-Carlo diagnostic of the conditional identities over the full corrected study. The MSE residual is empirical squared error minus the sample mean of $b(z)^2 + v(z)$; the bias residual is empirical error minus the sample mean of $b(z)$. Each supplied residual is scaled by its own Monte Carlo SE; the last column is the fraction outside its pointwise normal 95% interval. Groups share assignments across mechanisms/models and are dependent. These are descriptive multiple pointwise checks, not independent tests, multiplicity-adjusted rejections or proof that every realized residual is zero. Each listed group has finite SE. The 15 matched regular graph residuals in Figure 4 retain their actual MC bounds.

Scope	Residual	Groups	Max $\lvert r/\mathrm{MCSE}\rvert$	Outside95% fraction
All regression groups	MSE residual	5,712	4.14	0.06443
All regression groups	Bias residual	5,712	3.27	0.07283
Matched controls	MSE residual	612	3.3	0.03922
Matched controls	Bias residual	612	3.27	0.08333

Table 15: Exposure spread, assignment units and matched-regression error on the same five regular $n = 2000, d = 20$ graphs/partitions, $p = .5$, $\rho = 0$, linear response/fit. All cells use $\gamma = 1$ and 400 assignment/noise attempts per graph. I/C denote individual/cluster assignment. Each row is one graph/design; exposure SD is averaged over assignments and its supplied normal 95% MC bounds are shown; RMSE uses delta-method MC bounds in outcome units. All 400 estimates are finite and endpoint-estimable. Assignment units are 2000 nodes for I and the actual BFS cluster count K for C. f_{in} is the fixed partitions within-cluster edge fraction, repeated for I to identify the same partition, not an individual-assignment property. Wider exposure and fewer independent units occur together; the comparison does not isolate their separate causal effects. Streams across designs are independent. No graph averaging is performed in this table.

Seed / design	Units	f_{in}	Exposure SD [MC95%]	RMSE [MC95%]
201 / I	2000	0.107	0.11112 [0.11094, 0.11129]	0.209 [0.197, 0.222]
201 / C	97	0.107	0.13522 [0.13501, 0.13543]	0.168 [0.157, 0.179]
202 / I	2000	0.107	0.11132 [0.11115, 0.1115]	0.204 [0.191, 0.218]
202 / C	97	0.107	0.13439 [0.13415, 0.13463]	0.164 [0.153, 0.176]
203 / I	2000	0.107	0.11106 [0.11089, 0.11123]	0.208 [0.194, 0.223]
203 / C	111	0.107	0.13504 [0.13479, 0.1353]	0.167 [0.156, 0.178]
204 / I	2000	0.107	0.11122 [0.11106, 0.11139]	0.214 [0.2, 0.228]
204 / C	111	0.107	0.13497 [0.13473, 0.1352]	0.157 [0.146, 0.169]
205 / I	2000	0.106	0.11119 [0.11103, 0.11136]	0.2 [0.187, 0.213]
205 / C	98	0.106	0.13433 [0.13409, 0.13457]	0.172 [0.16, 0.183]

Table 16: Independent-stream GATE RMSE contrasts, matched linear response/fit, $p = .5$, $\gamma = 1$. The C-minus-I block uses $\rho = 0$ and the noise block changes ρ at fixed assignment design. Five independent graph draws per synthetic regime; entries are equally weighted means of graph summaries with approximate 95% t_4 intervals, including finite Monte Carlo variation. SNAP is one fixed topology: its bounds are conditional Monte Carlo intervals, not graph-population intervals. Positive means the first named condition has larger RMSE, in outcome units. The analysis supplies independent-panel/design Monte Carlo contrasts before graph averaging; shared graph names do not imply paired assignment/noise replicates. All selected outputs are finite and endpoint-estimable, 400 per condition/graph. ER/SBM/BA are compound structural regimes, not isolated topology interventions.

Graph / design	Contrast	RMSE difference [95%]
Regular 2000/4	C minus I	-0.0331 [-0.0353, -0.0308]
Regular 2000/20	C minus I	-0.0416 [-0.0539, -0.0292]
Regular 2000/80	C minus I	-0.0496 [-0.0787, -0.0205]
Regular 8000/20	C minus I	-0.0176 [-0.0222, -0.0129]
ER 2k	C minus I	-0.0303 [-0.042, -0.0187]
SBM 2k	C minus I	-0.0651 [-0.0729, -0.0573]
BA 2k	C minus I	-0.029 [-0.0368, -0.0212]
SNAP (fixed)	C minus I	-0.0569 [-0.0654, -0.0484]
Regular 2k d20 / I	$\rho 1$ minus $\rho 0$	0.0212 [0.0163, 0.0262]
Regular 2k d20 / C	$\rho 1$ minus $\rho 0$	0.0243 [0.0121, 0.0365]
SBM 2k / I	$\rho 1$ minus $\rho 0$	0.028 [0.0161, 0.0399]
SBM 2k / C	$\rho 1$ minus $\rho 0$	0.053 [0.0442, 0.0619]
SNAP (fixed) / I	$\rho 1$ minus $\rho 0$	0.147 [0.127, 0.166]
SNAP (fixed) / C	$\rho 1$ minus $\rho 0$	0.0789 [0.0692, 0.0886]

Table 17: Negative-control and baseline scope on sparse regular $n = 2000, d = 4$ graphs, $p = .5$, $\rho = 0$. All cells use $\gamma = 1$ and 400 assignment/noise attempts per graph. I/C denote individual/cluster assignment. Five independent graph draws per synthetic regime; entries are equally weighted means of graph summaries with approximate 95% t_4 intervals, including finite Monte Carlo variation. Entries are operational GATE RMSE in outcome units and mean defined fractions f_D. Unnecessary exposure adjustment costs precision without interference; with linear spillover DIM misses the spillover effect. Hájek risk conditions on a defined return; its undefined outputs are never assigned zero error. No interval is shown for HT/Hájek empirical RMSE because finite runs may miss rare-event risk; exact HT risk is separately retained. A displayed $f_D = 1$ describes the saved attempts, not a probability-one guarantee. These controls do not establish a universal ranking.

Response / design	Estimator	Empirical RMSE [95%]	f_D
No interference / I	Mean difference	0.0498 [0.0486, 0.0509]	1
No interference / I	Linear OLS	0.101 [0.0989, 0.103]	1
No interference / I	Exposure HT	0.2613	1
No interference / I	Exposure Hájek	0.2008	1
No interference / C	Mean difference	0.0503 [0.0479, 0.0528]	1
No interference / C	Linear OLS	0.0678 [0.0646, 0.071]	1
No interference / C	Exposure HT	0.3942	1
No interference / C	Exposure Hájek	0.1217	1
Linear spillover / I	Mean difference	1 [1, 1.01]	1
Linear spillover / I	Linear OLS	0.101 [0.0989, 0.103]	1
Linear spillover / I	Exposure HT	0.3918	1
Linear spillover / I	Exposure Hájek	0.2008	1
Linear spillover / C	Mean difference	0.541 [0.536, 0.545]	1
Linear spillover / C	Linear OLS	0.0678 [0.0646, 0.071]	1
Linear spillover / C	Exposure HT	0.7591	1
Linear spillover / C	Exposure Hájek	0.1217	1

Table 18: Complete corrected-study estimator failure accounting across all 102 cells and eight mechanisms. Each estimator has 326,400 output rows = 40,800 assignment-cell attempts times eight responses. U counts undefined outputs; S counts finite outputs whose endpoint contrast is unsupported, retained in operational error. Event columns deduplicate the mechanism copies to distinct assignment-cell events; copied responses are not independent failure observations. No cell is failed or unrun. Zero observed failures do not assert probability zero. Hájek U is an empty-exposure ratio; spline/threshold S is not successful endpoint recovery. DIM/HT failure policies remain implemented even though their U counts are zero.

Estimator	U outputs	S outputs	U events	S events
Mean difference	0	0	0	0
Linear OLS	0	0	0	0
Quadratic OLS	0	0	0	0
Cubic OLS	0	0	0	0
Fixed spline	0	34,704	0	4,338
Two-hop OLS	0	0	0	0
Threshold oracle	0	23,672	0	2,959
Exposure HT	0	0	0	0
Exposure Hájek	159,080	0	19,885	0

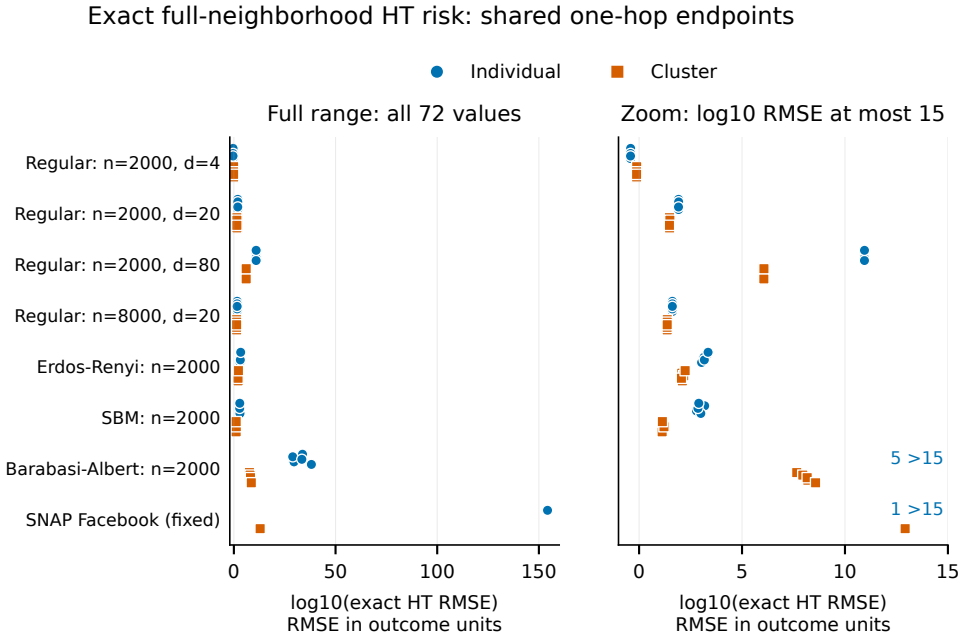


Figure 5: Exactly evaluated full-neighborhood exposure HT RMSE for all 36 core graphs under individual and cluster assignment ($p = 0.5$, $\rho = 0$, $\gamma = 1$). Each marker is one graph/design value; five synthetic seeds have fixed small vertical offsets, while SNAP is one fixed public topology. The full-range panel contains all 72 values. The zoom duplicates values at log10 RMSE at most 15; labels count values beyond the zoom, retained at their true positions in the overview. An x coordinate k means RMSE 10^k in outcome units. Values use the supplied exact joint-exposure variance calculation, integrated over assignment and Gaussian noise conditional on the graph/covariates. There are no Monte Carlo intervals or estimated graph-population intervals. Shown endpoints are shared by linear, quadratic, cubic, sigmoid and threshold one-hop responses; the companion table also reports no-interference and hidden-tail endpoints. The one-hop unbiased-risk identity does not apply to the two-hop response. This is HT, not the partially defined full-exposure Hájek ratio. Exact risk is distinct from finite-run empirical RMSE when rare endpoint events go unobserved; no empirical risk comparison is made here.

D.1 Exact one-hop HT risks across all planned cells

Figure 5 preserves the full core range with an explicitly counted duplicate zoom. Tables 19 and 20 retain all 306 endpoint-family values, including risks beyond the core figure's scope. These are conditional exact calculations under the assumptions of Appendix A.3; empirical finite-run HT errors are separately labeled in Table 7.

Table 19: Exactly evaluated full-neighborhood HT risk in log10 RMSE units, conditional on each core graph/covariates ($p = 0.5$, $\rho = 0$, $\gamma = 1$). I and C mean individual and cluster assignment. No-int., Shared and Tail are the no-interference, shared linear/quadratic/cubic/sigmoid/threshold one-hop, and hidden-tail endpoint families. A table entry k represents RMSE 10^k in outcome units. These exact assignment/noise risks have no Monte Carlo interval; each row is a separate graph, not a graph-family mean. The one-hop unbiased-risk identity excludes the two-hop response; no Hájek risk or finite-run performance ranking is asserted. Entries are rounded to two decimals; full precision is retained in the source ledger. Reg. is regular; ER, SBM and BA are Erdos–Renyi, stochastic-block and Barabasi–Albert graphs. n2k/n8k mean 2000/8000 nodes; d is regular degree.

Graph	Seed	No-int. I	No-int. C	Shared I	Shared C	Tail I	Tail C
Reg. n2k d4	201	-0.58	-0.40	-0.41	-0.12	-0.27	0.06
Reg. n2k d4	202	-0.58	-0.38	-0.41	-0.11	-0.27	0.06
Reg. n2k d4	203	-0.59	-0.41	-0.42	-0.12	-0.27	0.05
Reg. n2k d4	204	-0.59	-0.41	-0.42	-0.12	-0.27	0.05
Reg. n2k d4	205	-0.59	-0.41	-0.41	-0.12	-0.27	0.05
Reg. n2k d20	201	1.78	1.32	1.92	1.48	2.04	1.61
Reg. n2k d20	202	1.79	1.34	1.92	1.50	2.04	1.63
Reg. n2k d20	203	1.78	1.31	1.91	1.47	2.04	1.61
Reg. n2k d20	204	1.78	1.32	1.91	1.47	2.04	1.61
Reg. n2k d20	205	1.78	1.33	1.92	1.48	2.04	1.62
Reg. n2k d80	201	10.81	5.90	10.95	6.06	11.07	6.20
Reg. n2k d80	202	10.82	5.92	10.95	6.08	11.08	6.22
Reg. n2k d80	203	10.81	5.90	10.95	6.07	11.07	6.21
Reg. n2k d80	204	10.81	5.88	10.94	6.04	11.07	6.18
Reg. n2k d80	205	10.81	5.90	10.95	6.06	11.07	6.20
Reg. n8k d20	201	1.48	1.23	1.62	1.37	1.74	1.49
Reg. n8k d20	202	1.48	1.23	1.62	1.37	1.74	1.49
Reg. n8k d20	203	1.48	1.23	1.62	1.37	1.74	1.49
Reg. n8k d20	204	1.48	1.23	1.61	1.36	1.74	1.49
Reg. n8k d20	205	1.48	1.23	1.62	1.37	1.74	1.49
ER n2k	201	3.21	2.10	3.35	2.24	3.47	2.36
ER n2k	202	3.14	1.94	3.29	2.07	3.42	2.20
ER n2k	203	3.01	2.02	3.15	2.17	3.28	2.29
ER n2k	204	3.04	1.91	3.17	2.06	3.30	2.19
ER n2k	205	2.91	1.96	3.04	2.10	3.16	2.23
SBM n2k	201	2.76	0.94	2.90	1.13	3.02	1.28
SBM n2k	202	3.04	0.96	3.19	1.15	3.32	1.30
SBM n2k	203	2.72	1.02	2.86	1.21	2.99	1.36
SBM n2k	204	2.66	0.99	2.79	1.17	2.91	1.32
SBM n2k	205	2.87	0.92	3.00	1.12	3.13	1.27
BA n2k	201	33.62	7.51	33.75	7.66	33.89	7.79
BA n2k	202	28.81	7.84	28.94	7.95	29.08	8.08
BA n2k	203	33.38	8.05	33.43	8.16	33.54	8.30
BA n2k	204	29.44	8.05	29.58	8.19	29.69	8.32
BA n2k	205	37.99	8.43	38.12	8.57	38.26	8.70
SNAP Facebook	201	154.11	12.78	154.26	12.92	154.38	13.05

Table 20: Exact one-hop full-neighborhood HT risks for all 30 noise/imbalance cells beyond the 72 core cells in Table 19. Entries are $\log_{10}$(RMSE): k means RMSE 10^k in outcome units. No-int./Shared/Tail are no-interference, shared one-hop endpoint, and hidden-tail families. I/C denote individual/cluster assignment; $\gamma = 1$. Reg./SBM rows identify the graph seed (regular $n = 2000, d = 20$; SBM $n = 2000$); SNAP is the fixed full topology. These are graph/covariate-conditional exact assignment/noise calculations with no Monte Carlo interval or graph-population claim. The one-hop unbiased-risk identity excludes the two-hop response, and no Hájek identity is implied. The source ledger retains all 306 endpoint-family rows.

Graph/seed	Design	p	ρ	No-int.	Shared	Tail
Reg. 201	I	0.2	0	5.866	6.050	6.195
Reg. 201	I	0.8	0	5.735	5.735	5.735
Reg. 201	C	0.2	0	4.970	5.153	5.298
Reg. 201	C	0.8	0	4.841	4.841	4.841
SNAP	I	0.2	0	362.156	362.344	362.485
SNAP	I	0.8	0	361.967	361.967	361.967
SNAP	C	0.2	0	33.990	34.178	34.318
SNAP	C	0.8	0	33.800	33.800	33.800
Reg. 201	I	0.5	1	1.782	1.917	2.041
Reg. 201	C	0.5	1	1.320	1.477	1.611
Reg. 202	I	0.5	1	1.788	1.921	2.044
Reg. 202	C	0.5	1	1.341	1.495	1.628
Reg. 203	I	0.5	1	1.780	1.914	2.039
Reg. 203	C	0.5	1	1.315	1.471	1.606
Reg. 204	I	0.5	1	1.779	1.914	2.038
Reg. 204	C	0.5	1	1.318	1.474	1.609
Reg. 205	I	0.5	1	1.782	1.916	2.040
Reg. 205	C	0.5	1	1.325	1.481	1.616
SBM 201	I	0.5	1	2.763	2.898	3.020
SBM 201	C	0.5	1	0.940	1.131	1.278
SBM 202	I	0.5	1	3.039	3.186	3.316
SBM 202	C	0.5	1	0.961	1.152	1.300
SBM 203	I	0.5	1	2.721	2.858	2.987
SBM 203	C	0.5	1	1.025	1.209	1.355
SBM 204	I	0.5	1	2.657	2.786	2.910
SBM 204	C	0.5	1	0.991	1.174	1.320
SBM 205	I	0.5	1	2.866	2.998	3.130
SBM 205	C	0.5	1	0.926	1.122	1.273
SNAP	I	0.5	1	154.109	154.257	154.381
SNAP	C	0.5	1	12.776	12.923	13.048

E Research workflow and asset provenance

The study was developed by language-model agents assigned to research context, study planning, independent feasibility, analysis, presentation, venue verification, manuscript integration, and independent review. Their work includes methodological discussion, code construction, source consultation, and manuscript writing. The estimator uses no language model, and its outcomes are generated by the stated numerical mechanisms. Language-model assistance is part of the research workflow and supplies no independent empirical evidence. Executable checks, preserved outputs, source manifests, and review of the exact delivery are needed to assess the resulting work.

Only exact permitted starter-source and data-source locations were used for this manuscript's related-work and graph-attribution consultation; the direct reading record is `notes/source-reading.md`. A requested original graph-source citation was retrieved from the official NIPS 2012 proceedings after the obsolete author-site link failed. This position encountered no known prohibited source. That statement is limited to its own consultation; the analysis and context positions retain their own source-access records. No title, author, or result of the reserved study is inferred here.

The SNAP topology is attributed to McAuley and Leskovec [2012] and the actual download page [Stanford Network Analysis Project]. The original paper's ego-network modeling convention excludes the ego, whereas the separately supplied combined download includes ego nodes. We use the combined download without imposing that modeling convention on it. Acquisition bytes are verified against the frozen SHA-256, and raw data are downloaded from the source rather than redistributed with an invented license. The supplied page states no data license. The original paper is used for attribution and historical collection context; its circle-detection results are not GATE evidence. The unchanged official NeurIPS 2026 formatting assets have separate preserved source hashes under `venue/`.

The delivered code package records Python and numerical-library versions, graph acquisition and processing, configurations, seed streams, failure policies, and reproduction commands. Source installation, scientific checks and the complete launcher are documented in the root `README.md` and `analysis/delivery_reproduction.md`. Appendix C reports the corrected selected replay, the superseded implementation defect and the measured execution limits. The complete corrected execution and validity receipts are supplied in `analysis/final_execution.json`; source-matched presentation manifests document displayed assets. This report records no new human-participant recruitment or institutional approval.

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